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A Appendix

A.1 Bounds on the Procrustes distance in terms of the Euclidean distance

Here we derive upper and lower bounds on the Procrustes distance between neural representations $\boldsymbol{X} \in \mathbb{R}^{M \times N_X}$ and $\boldsymbol{Y} \in \mathbb{R}^{M \times N_Y}$ in terms of the Euclidean distance between linear kernel matrices $\boldsymbol{K}_X = \boldsymbol{X}\boldsymbol{X}^\top$ and $\boldsymbol{K}_Y = \boldsymbol{Y}\boldsymbol{Y}^\top$. This result is applied in the main text to relate the expected Euclidean distance between decoded signals $\mathbb{E}\|\boldsymbol{X}\boldsymbol{w}^* - \boldsymbol{Y}\boldsymbol{v}^*\|_2^2 = \|\tilde{\boldsymbol{X}}\tilde{\boldsymbol{X}}^\top - \tilde{\boldsymbol{Y}}\tilde{\boldsymbol{Y}}^\top\|_F^2$ to the Procrustes distance $\mathcal{P}(\tilde{\boldsymbol{X}}, \tilde{\boldsymbol{Y}})$. Tildes are omitted in this section for clarity, but we note that this result is applied in the main text to the normalized representations $\tilde{\boldsymbol{X}}$ and $\tilde{\boldsymbol{Y}}$.

We will make use of the equivalence between the Procrustes distance between representation matrices $\boldsymbol{X}$ and $\boldsymbol{Y}$ and a notion of distance on the space of positive semi-definite kernel matrices $\boldsymbol{K}_X = \boldsymbol{X}\boldsymbol{X}^\top$ and $\boldsymbol{K}_Y = \boldsymbol{Y}\boldsymbol{Y}^\top$ called the Bures distance, defined as

$$d_B(\boldsymbol{K}_X, \boldsymbol{K}_Y) = \sqrt{\mathrm{Tr}[\boldsymbol{K}_X] + \mathrm{Tr}[\boldsymbol{K}_Y] - 2\,\mathrm{Tr}\left[\left(\boldsymbol{K}_X^{1/2}\boldsymbol{K}_Y\boldsymbol{K}_X^{1/2}\right)^{1/2}\right]}. \tag{27}$$

The third trace term on the right hand side of eq. (27) is often called the *fidelity*, defined as

$$\mathcal{F}(\boldsymbol{K}_X, \boldsymbol{K}_Y) = \mathrm{Tr}\left[\left(\boldsymbol{K}_X^{1/2}\boldsymbol{K}_Y\boldsymbol{K}_X^{1/2}\right)^{1/2}\right]. \tag{28}$$

A.1.1 Lower bound

For the lower bound, we are inspired by the Fuchs-van-de-Graaf inequalites that are used in quantum information theory to assert an approximate equivalence between two measures of quantum state similarity, the *trace distance* and the *fidelity*. This approach is relevant from a mathematical perspective for our purposes, since quantum states are represented by positive semi-definite matrices normalized to have trace 1. Here, we use a similar approach as that often used to derive the Fuchs-van de Graaf inequalites to instead relate the Euclidean distance and the Bures distance on positive semidefinite matrices, while also relaxing the trace 1 normalization.

We will make use of the following identity.

Lemma 1.

$$\|\alpha uu^\dagger - \beta vv^\dagger\|_* = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta|\langle u, v\rangle|^2} \tag{29}$$

for all unit vectors u, v and all non-negative real numbers α and β.

Proof. First we recognize that the nuclear norm of a matrix can be calculated by summing the singular values of that matrix. Furthermore, if that matrix is Hermitian, the singular values are the absolute values of the eigenvalues. Define $\boldsymbol{M} = \alpha uu^\dagger - \beta vv^\dagger$. Our matrix M is Hermitian and has at most two eigenvalues, so we will look for expressions for these in terms of α and β. Note that if u and v are the same unit vector, then $\boldsymbol{M}$ has rank 1, so we expect the eigenvalues will depend also on the inner product $\langle u, v\rangle$.

The eigenvectors of $\boldsymbol{M}$ will be of the form $\psi = cu + dv$ for some scalar c and d. By direct calculation:

$$\begin{aligned} \boldsymbol{M}\psi &= \lambda\psi \\ \boldsymbol{M}(cu + dv) &= (\alpha c + \alpha d\langle u, v\rangle)u - (\beta c\langle v, u\rangle + \beta d)v \\ &= \lambda(cu + dv) \end{aligned} \tag{30}$$

Therefore, we see that the eigenvalues must satisfy both

$$\lambda = \frac{\alpha c + \alpha d\langle u, v\rangle}{c} \quad \text{and} \quad \lambda = -\frac{\beta c\langle v, u\rangle + \beta d}{d} \tag{31}$$

Setting these equal to each other and solving the resulting quadratic for c gives: